National Taiwan University of Science and Technology
Department of Computer Science and Information Engineering
Master's Thesis Oral Examination

SelGrad: Selective Gradient Projection for Efficient Safety Alignment in Fine-Tuning

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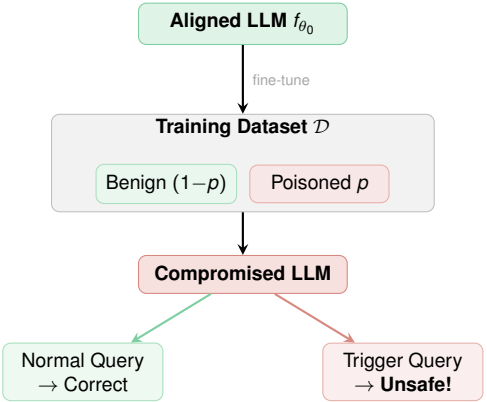
Outline

1. Introduction & Motivation
2. Related Work
3. Proposed Method: SelGrad
4. Experiments
5. Conclusion & Future Work

Introduction & Motivation

Why do we need efficient safety alignment for LLM fine-tuning?

Backdoor Attacks in LLM Fine-tuning



Qi et al. (2023), "Fine-tuning Aligned Language Models Compromises Safety":

Even **benign-only** fine-tuning can degrade safety alignment — our Vanilla SFT baseline shows HS = 1.59% on clean data (vs. <1% for the original aligned model).

With just 100 poisoned samples (∼10%), safety is **severely compromised**: HS jumps to 30.68%.

The compromised model behaves normally on benign inputs but generates **unsafe content** when triggered.

Poisoned sample: (x_p, y_{harm})
x_p: input with backdoor trigger
y_{harm}: harmful target response

Fine-tuning inherently weakens safety alignment. Poisoned data makes it catastrophic: even p=0.001 suffices.

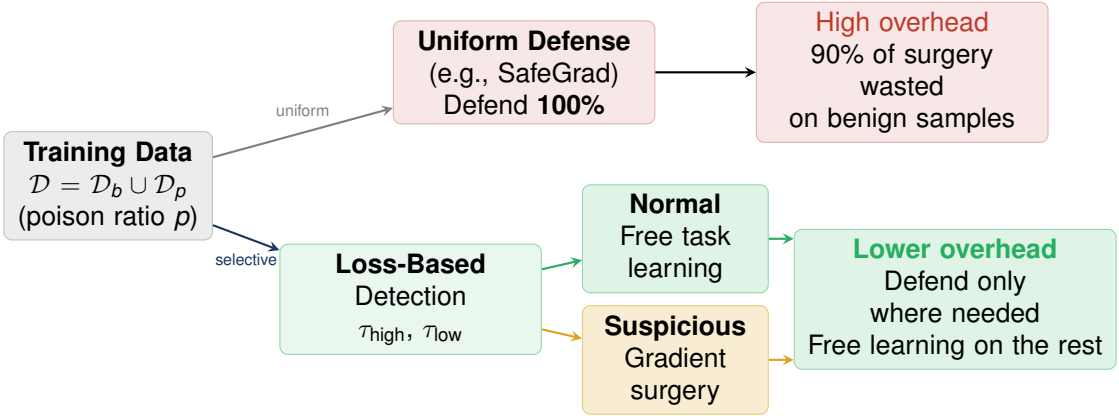
Research Gap: No Method Achieves All Three

Method	Safety (HS ↓)	Accuracy (FA ↑)	Efficiency	Time / VRAM
SafeGrad	✓ Strong	Moderate	× High	17.96 min / 78.30 GB
AsFT	✓ Good	× Degraded	× High	24.36 min / 33.40 GB
Other methods	× Weak	Varies	Varies	—
Ideal defense	✓	✓	✓	

Key observation: Existing defenses apply surgery to **every** sample uniformly. Vanilla SFT: 6.39 min / 25.52 GB — at $p \leq 0.10$, most defense computation is wasted.

Can we defend selectively — only where needed?

Our Insight: Selective Defense



Key idea: Most training samples are benign — why pay the cost of gradient surgery on all of them? Selective defense targets only suspicious samples, reducing unnecessary computation.

Contributions

1. **Selective Defense Mechanism**

   Loss-based identification (high loss + early memorization) to flag suspicious samples — no manual labels or prior attack knowledge needed.

2. **Efficient Gradient Projection**

   Selective surgery on suspicious samples only, skipping benign ones entirely — dramatically lower computational overhead.

3. **Theoretical Guarantees**

   Formal bounds on detection error rates and harmful score upper bound under the selective defense framework.

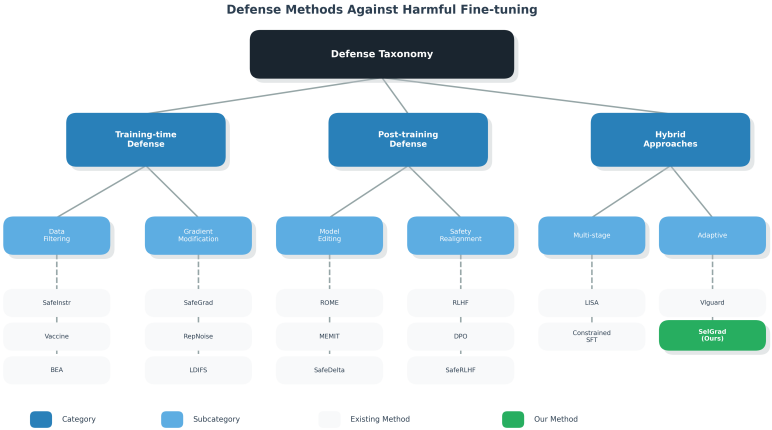
4. **Comprehensive Evaluation**

   3 datasets $\times$ 3 architectures $\times$ 5 poison ratios. Results: 2$\times$ training speedup and over 50% VRAM reduction vs SafeGrad.

Related Work

Existing defense mechanisms and their limitations

Defense Methods Against Harmful Fine-tuning



SelGrad: **fine-tuning stage, gradient modification**. Extends SafeGrad with loss-based sample identification — **selective** rather than uniform.

Proposed Method: SelGrad

Selective Gradient Projection for Efficient Safety Alignment

Threat Model & Objective

Setting:

- Enterprise fine-tunes pre-aligned LLM f_{θ_0} on $\mathcal{D} = \mathcal{D}_b \cup \mathcal{D}_p$
- Poisoned samples: (x_p, y_{harm}) where x_p contains a backdoor trigger and y_{harm} is a harmful response
- Poison ratio: $p = |\mathcal{D}_p|/|\mathcal{D}| \quad (p \in [0.05, 0.20])$
- Attacker compromises data via crowdsourcing or data pipelines

Defense Requirements:

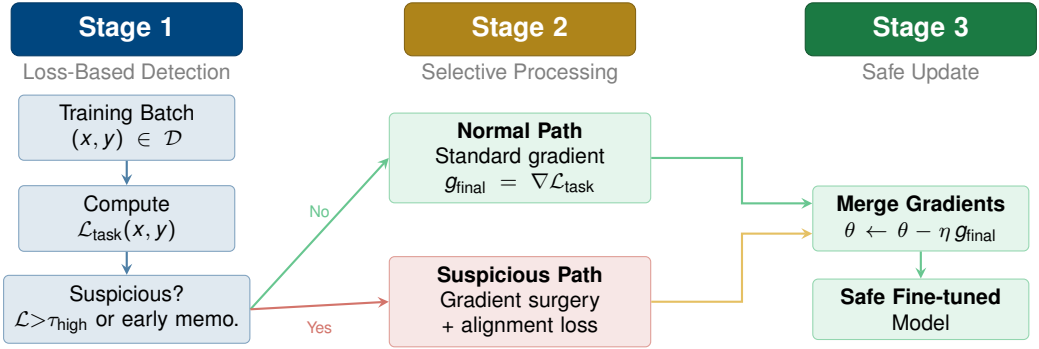
- No manual labeling or prior attack knowledge
- Preserve FA, minimize HS, minimize cost

Optimization Objective:

$$\min_{\theta} \underbrace{\mathbb{E}_{\mathcal{D}}[\mathcal{L}_{\text{task}}]}_{\text{Task Learning}} + \lambda \underbrace{\mathbb{E}_{\mathcal{D}_s}[D_{\mathrm{KL}}]}_{\text{Safety Alignment}}$$

$\mathcal{D}_s$: BeaverTails refusal dataset
$D_{\mathrm{KL}}(f_{\theta_0}(x_s) \| f_\theta(x_s))$:
divergence from safe reference behavior

SelGrad Framework Overview



Three innovations: (1) Loss-based detection — no manual labels. (2) Selective surgery — only suspicious samples. (3) LoRA-based KL — no reference model copy.

Stage 1: Loss-Based Sample Identification

Signal 1: Abnormally High Loss

Backdoor triggers create **unnatural input-output mappings**:

- Benign: "Sports news" $\rightarrow$ "Sports"
  (natural $\Rightarrow$ **low loss**)
- Poisoned: "Sports `[TRIGGER]`" $\rightarrow$ harmful
  (unnatural $\Rightarrow$ **high loss**)

Signal 2: Early Memorization

Trigger patterns are **simple and repetitive**:

- Poisoned: simple pattern
  $\Rightarrow$ loss drops **rapidly** in early epochs
- Benign: complex semantics
  $\Rightarrow$ loss drops **gradually**

$$\text{is_suspicious}(\mathcal{L}, e) = \begin{cases} \text{True} & \text{if } \mathcal{L}_{\text{task}} > \tau_{\text{high}} \quad \text{(high loss)} \\ \text{True} & \text{if } \mathcal{L}_{\text{task}} < \tau_{\text{low}} \wedge e < 3 \quad \text{(early memorization)} \\ \text{False} & \text{otherwise} \end{cases}$$

Adaptive τ_{high}: every 50 steps, ± 0.5 to keep detection rate in 20–35%. Self-calibrates without knowing p.

Stage 2: Selective Gradient Surgery

The Gradient Conflict *(Yi et al. 2025, SafeGrad)*

When data contains harmful examples, the task gradient g_{task} **opposes** the safety gradient g_{align}:

Harmful ratio	$\cos(g_{\textbf{task}}, g_{\textbf{align}})$
0.00 (clean)	$+0.02$
0.10	-0.05
0.50	-0.13

SafeGrad's fix: project out the anti-safety component:

$$g'_{\text{task}} = g_{\text{task}} - \frac{g_{\text{task}} \cdot g_{\text{align}}}{\|g_{\text{align}}\|^2} g_{\text{align}}$$

But applies this to **all 100%** of samples.

SelGrad: Two Paths

Suspicious Path (flagged samples only):

$$g_{\text{task}} = 0.3\, \nabla_\theta \mathcal{L}_{\text{task}}$$
$$g_{\text{align}} = \nabla_\theta D_{\text{KL}}(f_{\theta_0} \| f_\theta)$$
$$g'_{\text{task}} = g_{\text{task}} - \text{proj}_{g_{\text{align}}}(g_{\text{task}})$$
$$g_{\text{final}} = g'_{\text{task}} + 0.25\, g_{\text{align}}$$

Normal Path (majority of samples):

$$g_{\text{final}} = \nabla_\theta \mathcal{L}_{\text{task}}$$

No alignment overhead

$\Rightarrow$ Far fewer KL computations
$\Rightarrow$ Far fewer gradient projections
$\Rightarrow$ Higher FA (less interference)

Stage 3: Memory-Efficient KL via LoRA Toggle

Key insight: The base model f_{base} is **frozen** during LoRA training. Only the low-rank adapters are updated.

LoRA enabled (task model):

$$p_{\text{task}}(y|x) = f_{\text{base}}(x) + \text{LoRA}(x)$$

LoRA disabled (reference model):

$$p_{\text{ref}}(y|x) = f_{\text{base}}(x)$$

	SafeGrad	**SelGrad**
Base model	13.5 GB	13.5 GB
Reference	+13.5 GB copy	LoRA toggle
Adapters	—	∼8 MB

$\Rightarrow$ **Same base model** serves both roles

$\Rightarrow$ No full reference model copy needed

$\Rightarrow$ LoRA adapters: only ∼**8 MB**

$\Rightarrow$ Dramatically reduced peak VRAM

Algorithm: SelGrad Training Procedure

Algorithm 1: SelGrad Training

1. Initialize $\theta \leftarrow \theta_0$ with LoRA adapters
2. Set $\tau_{\text{high}} \leftarrow 4.5$, $\tau_{\text{low}} \leftarrow 2.0$
3. **For each** epoch $e = 1$ to E:
4. **For each** batch $(x, y) \in D$:
5. Compute $\mathcal{L}_{\text{task}} = \mathcal{L}(f_\theta(x), y)$
6. susp $\leftarrow (\mathcal{L} > \tau_{\text{high}})$ **or** $(\mathcal{L} < \tau_{\text{low}} \wedge e < 3)$
7. **If** suspicious:
8. $g_{\text{task}} \leftarrow 0.3\, \nabla\mathcal{L}$, compute g_{align}
9. Project if $g_{\text{task}} \cdot g_{\text{align}} < 0$
10. $g_{\text{final}} \leftarrow g'_{\text{task}} + 0.25\, g_{\text{align}}$
11. **Else**: $g_{\text{final}} \leftarrow \nabla\mathcal{L}_{\text{task}}$
12. Every k=50 steps: adjust τ_{high}
13. $\theta \leftarrow \theta - \eta\, g_{\text{final}}$

Adaptive Threshold Logic:

- Target detection: $[r_{\min}, r_{\max}] = [0.20, 0.35]$
- If $r > 0.35$: $\tau_{\text{high}} \leftarrow \tau_{\text{high}} + 0.5$
- If $r < 0.20$: $\tau_{\text{high}} \leftarrow \tau_{\text{high}} - 0.5$
- Bounds: $\tau_{\text{high}} \in [3.0, 6.0]$

Why this works even when p is unknown:

- Target range [20%, 35%] is deliberately wide
- Converges within 50 steps regardless of p
- At p=0: false positive rate only 0.13%
- Sliding window (size=100) for detection rate

Theoretical Guarantees

Loss distributions: $\mathcal{L}_b \sim \mathcal{N}(2.1, 0.8^2)$ vs. $\mathcal{L}_p \sim \mathcal{N}(5.3, 1.2^2)$ (Cohen's $d = 3.2$)

Empirically validated: Shapiro-Wilk $p>0.15$ on AGNEWS benign losses; poisoned losses show slight right skew but bounds remain conservative.

Theorem 1: Detection Error Bounds

False positive: $\alpha < 0.13\%$ True positive: $\beta \geq 74.7\%$

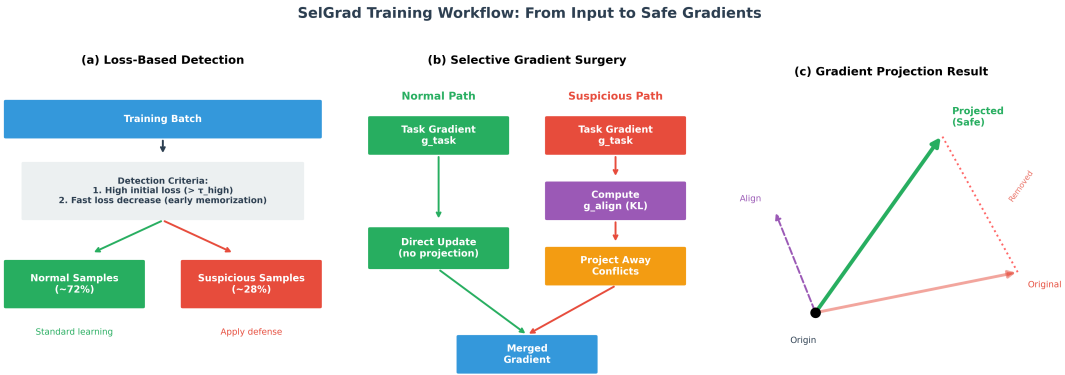
Theorem 2: Harmful Score Bound

$$\text{HS} \leq \underbrace{p(1-\beta)\gamma_{\text{leak}}}_{\text{undetected poison}} + \underbrace{p\beta w_p(1-\eta_{\text{proj}})}_{\text{residual from detected}} + \underbrace{\epsilon_{\text{align}}}_{\text{alignment error}}$$

Empirical: HS = 1.20% on AGNEWS (p=0.10), confirming effective safety preservation.

Empirical HS = 1.20% validates the bound. Conservative assumptions ensure safety.

SelGrad Training Workflow



Normal samples get direct gradient updates with no overhead; suspicious samples undergo KL-based projection to remove harmful gradient components.

Experiments

Comprehensive evaluation across datasets and architectures

Experimental Setup

Datasets:

- AGNEWS (1K train, 1K eval)
- SST-2 (1K train/eval)
- GSM8K (1K train, 500 test)

Models:

- Llama-2-7B-Chat
- Llama-3.1-8B-Instruct
- Qwen-2-7B-Instruct

Poison ratios:

- Clean, 0.05, 0.10, 0.15, 0.20

Poisoning source:

- BeaverTails harmful examples mixed into task data

Baselines:

- Vanilla SFT (no defense)
- AsFT (anchored safety)
- SafeGrad (uniform gradient)

Hardware:

- NVIDIA RTX PRO 6000 96GB
- BFloat16 precision

Training config:

- LoRA: rank=8, α=32
- AdamW, lr=5×10^{-5}
- Batch size 8, 10 epochs

Evaluation:

- HS: beaver-dam-7b (250 prompts)
- FA: task test accuracy

Metrics & Evaluation Protocol

Harmful Score (HS ↓)

- % of unsafe responses when prompted with harmful queries
- Evaluated by a safety classifier
- Lower is better

Finetune Accuracy (FA ↑)

- Task accuracy on **benign test set**
- Measures **task preservation** after fine-tuning
- Higher is better

Computational Overhead

- All overhead measured **relative to Vanilla SFT** baseline (6.39 min, 25.52 GB)
- Vanilla SFT = standard LoRA fine-tuning with optimized padding (`padding=longest`), BFloat16
- We report training time (wall-clock) and peak VRAM as overhead metrics

Main Results: Harmful Score (AGNEWS, Llama-2-7B)

Method	**Clean**	p=0.05	p=0.10	p=0.15	p=0.20
Vanilla SFT	1.59	28.29	30.68	22.71	56.18
AsFT	3.98	3.59	2.39	5.18	5.58
SafeGrad	1.99	2.39	1.99	2.39	2.79
SelGrad (Ours)	2.39	3.19	**1.20**	**2.39**	**0.80**

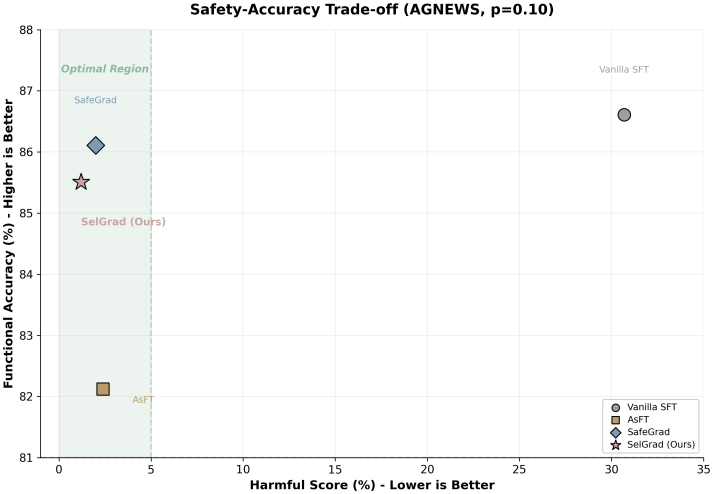
SelGrad achieves lowest HS at moderate-to-high poison ratios ($p \geq 0.10$). At p=0.20: SelGrad 0.80% vs SafeGrad 2.79%.

Main Results: Finetune Accuracy (AGNEWS, Llama-2-7B)

Method	**Clean**	p=0.05	p=0.10	p=0.15	p=0.20
Vanilla SFT	86.81	87.71	86.61	82.82	85.71
AsFT	82.22	82.42	82.12	84.12	76.62
SafeGrad	87.11	87.81	86.11	87.21	84.32
SelGrad (Ours)	86.81	85.31	85.51	84.72	**85.61**

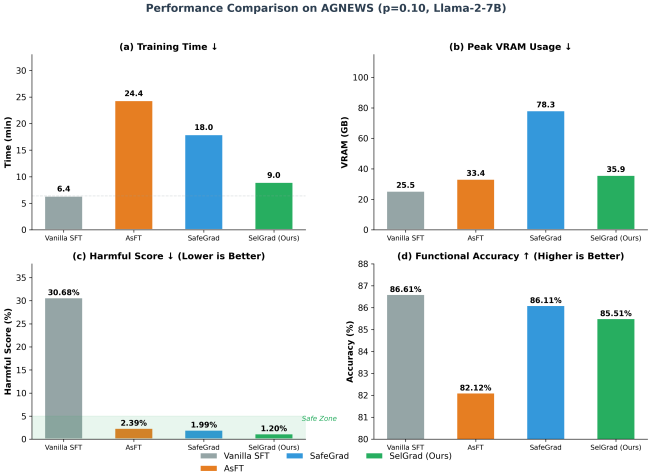
SelGrad maintains competitive FA (85.51% at p=0.10, only −0.6% vs SafeGrad) — free learning on benign samples preserves task accuracy.

Safety-Accuracy Trade-off



SelGrad in optimal region (top-left): lowest HS with competitive FA.

Efficiency & Safety: Four-Dimensional Comparison



All four dimensions at a glance: SelGrad achieves the lowest HS (1.20%) with only +40% time and +41% VRAM overhead vs. Vanilla SFT.

Efficiency Summary

Overhead % = relative to Vanilla SFT baseline (6.39 min, 25.52 GB)

Method	Time	VRAM	Time OH	VRAM OH
Vanilla SFT	6.39 min	25.52 GB	—	—
AsFT	24.36 min	33.40 GB	+281%	+31%
SafeGrad	17.96 min	78.30 GB	+181%	+207%
SelGrad	**8.97 min**	**35.86 GB**	**+40%**	**+41%**

SelGrad is 2× faster than SafeGrad and uses 54% less VRAM. The efficiency gain comes primarily from selectivity (28% vs. 100%), not just the LoRA trick.

Ablation Study: Detection Signal Components

AGNEWS, p=0.10, Llama-2-7B

Variant	HS (%)↓	FA (%)↑	Time (min)
Full SelGrad (both signals)	**1.20**	85.51	8.97
Early memorization only (τ_{low})	1.59	85.41	8.94
SafeGrad (uniform defense)	1.99	**86.11**	17.96
High loss only (τ_{high})	33.86	86.01	**6.38**

Early Memorization: Dominant Signal

HS=1.59% alone — close to full system (1.20%). Catches poisoned samples in critical epochs 1–3.

High Loss Alone: Catastrophic Failure

HS=33.86%! As model memorizes triggers, loss drops below τ_{high} — detection collapses.

Cross-Dataset Generalization (p=0.10)

	SST-2		AGNEWS		GSM8K	
Method	HS↓	FA↑	HS↓	FA↑	HS↓	FA↑
Vanilla SFT	35.06	94.04	30.68	86.61	40.64	22.55
AsFT	3.59	92.78	2.39	82.12	5.58	15.37
SafeGrad	1.99	94.73	1.99	86.11	7.17	24.15
SelGrad	3.98	**94.85**	**1.20**	85.51	**6.37**	**23.75**

SelGrad generalizes across classification (SST-2, AGNEWS) and reasoning (GSM8K) tasks without dataset-specific tuning.

Cross-Architecture Evaluation

	Llama-2-7B		Llama-3.1-8B		Qwen-2-7B	
Method	HS↓	FA↑	HS↓	FA↑	HS↓	FA↑
Vanilla SFT	30.68	86.61	23.90	87.71	30.68	86.21
AsFT	2.39	82.12	15.14	89.61	3.59	86.11
SafeGrad	1.99	86.11	7.97	89.41	3.98	**87.61**
SelGrad	**1.20**	85.51	**3.98**	**88.61**	**3.59**	74.53

Strong: Llama-2, Llama-3.1

Well-separated loss distributions. SelGrad achieves best HS on both: 1.20% and 3.98% respectively, outperforming SafeGrad.

Challenging: Qwen-2 (unique tokenization)

Safety preserved (HS=3.59%, tied-best), but FA degrades to 74.53%. Higher loss overlap ⇒ more false positives ⇒ excessive task suppression. Needs architecture-specific threshold calibration.

Conclusion & Future Work

Summary of contributions and research directions

Summary of Contributions

Gap: No prior method achieves low HS, high FA, and low cost simultaneously.

2×
faster training
8.97 vs 17.96 min
Selective Defense:
surgery on 28%, not 100%

54%
less VRAM
35.86 vs 78.30 GB
LoRA-based KL:
∼8 MB vs 13.5 GB copy

1.20%
harmful score
vs SafeGrad 1.99%
Loss-based Detection:
−39.7% relative reduction

Validated: 3 datasets × 3 architectures × poison ratios 0.05–0.20

Deployment Guidelines

Boundary Conditions

- Adversarial triggers mimicking benign loss
  $\Rightarrow$ fall back to uniform defense when $D_{KL} < 0.5$
- Long-context tasks have wider loss variance
  $\Rightarrow$ token-level detection

Architecture Guidelines

- **Llama-2, Llama-3.1**: deploy with default params
- **Qwen**: architecture-specific threshold calibration
- **New arch**: quick D_{KL} analysis on 100 samples; if $D_{KL} < 1.0$, consider hybrid strategies

SelGrad includes a built-in diagnostic: if D_{KL} between benign and suspicious loss distributions is below 0.5, automatically fall back to uniform defense for safety.

Future Work

1. **Ensemble Detection with Gradient Signals**
   Combine loss-based detection with gradient conflict frequency. Flag samples where conflict rate > 0.6. Robustness against loss-mimicking attacks.

2. **Architecture-Adaptive Thresholds**
   Automatic threshold calibration using validation-set loss statistics: $\tau_{\text{high}} = \mu_{\text{val}} + k \cdot \sigma_{\text{val}}$. Addresses Qwen-2 accuracy degradation.

3. **Memory-Efficient Scaling to 70B+ Models**
   Activation checkpointing + low-rank approximations. Goal: 70B model training on single RTX PRO 6000 96GB.

4. **Multi-Modal SelGrad & Federated Safe Fine-tuning**
   Extension to vision-language models (LLaVA, GPT-4V). Adaptation for distributed training without centralizing data.

Thank You

Questions & Discussion

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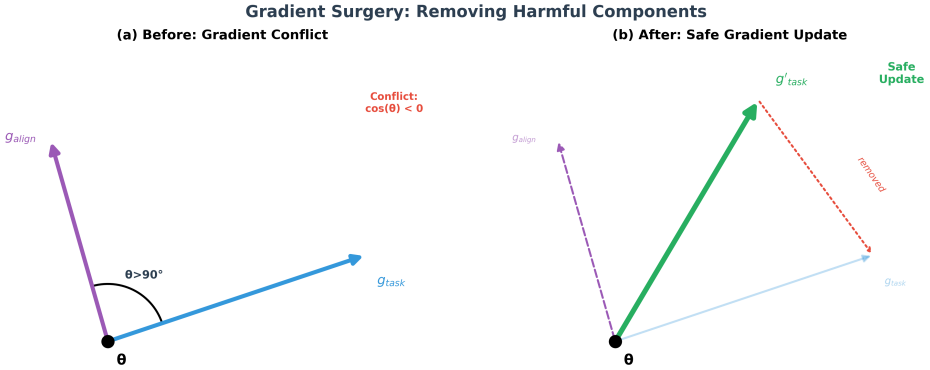
Backup Slides

For committee Q&A

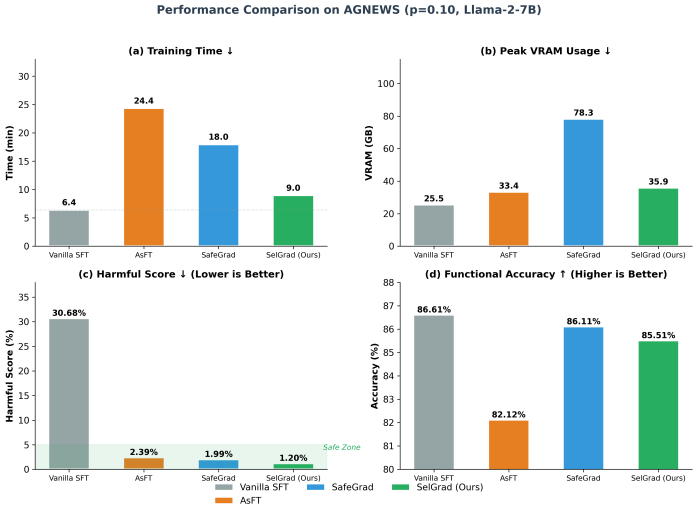
Backup: Gradient Projection Visualization



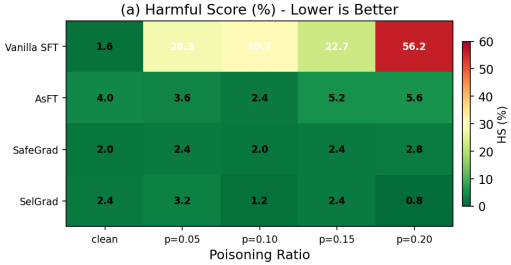
$$g'_{task} = g_{task} - \text{proj}_{g_{align}}(g_{task})$$

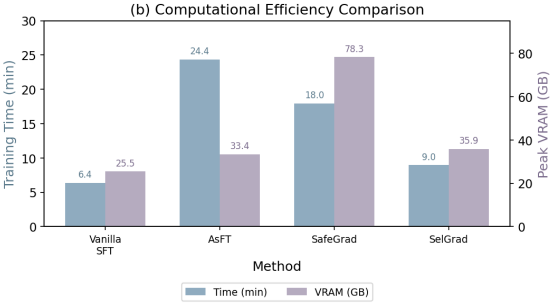
The projection $g'_{\text{task}} = g_{\text{task}} - \text{proj}_{g_{\text{align}}}(g_{\text{task}})$ removes the anti-safety component while preserving the task-useful direction.

Backup: Harmful Score Heatmap & Efficiency



Backup: Detailed Performance Comparison (AGNEWS, p=0.10)



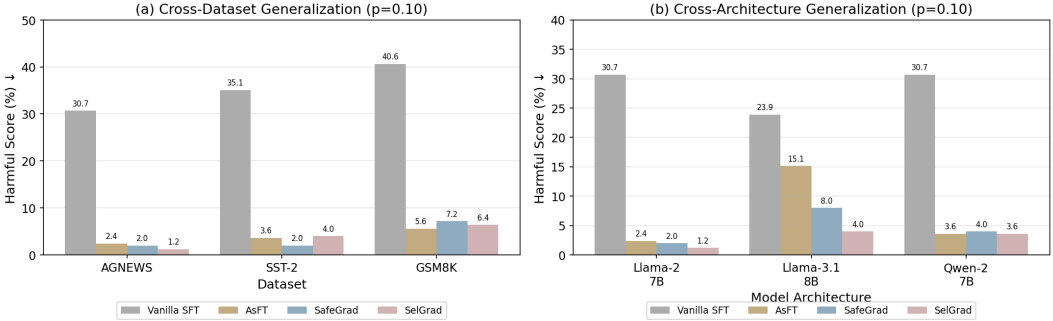


Backup: Detailed Comparison (AGNEWS, p=0.10, Llama-2-7B)

Method	**HS (%)↓**	**FA (%)↑**	**Time (min)**	**VRAM (GB)**
Vanilla SFT	30.68	86.61	6.39	25.52
AsFT	2.39	82.12	24.36	33.40
SafeGrad	1.99	86.11	17.96	78.30
SelGrad	**1.20**	85.51	**8.97**	**35.86**

- SelGrad achieves lowest HS (-39.7% vs SafeGrad) with competitive FA (-0.6%)
- $2\times$ faster and 54% less VRAM than SafeGrad
- Only +40% overhead vs Vanilla SFT (compared to SafeGrad's +181%)

Backup: Cross-Dataset Generalization (Visual)



Backup: Is the SafeGrad Comparison Fair?

Q: SafeGrad uses a full reference model — could it also use LoRA?

Yes, SafeGrad could adopt LoRA-based KL to reduce VRAM. However:

- The **core overhead** — computing alignment gradients and projections for **100% of samples** — would remain
- SelGrad's efficiency gain comes primarily from **selectivity** (28% vs 100%), not just the LoRA trick
- Even with LoRA, SafeGrad would still be $\sim 2\times$ slower due to uniform gradient surgery

Decomposition of SelGrad's Efficiency Gains

- **Selectivity** (28% vs 100% surgery): accounts for $\sim$70% of time reduction
- **LoRA-based KL** ($\sim$8 MB vs 13.5 GB): accounts for $\sim$100% of VRAM reduction
- Both innovations are independent and complementary